Measure Theoretic Probability
Lesson 26
General Expectation and Integration with Respect to a Measure

Kirschner Awonatey Amaanya

Department of Statistics and Actuarial Science, KNUST

Outline

1. Recap and Plan
2. Expectation of a General Random Variable
3. Integration with Respect to a Measure
4. Four Basic Facts
5. Summary and Exercises

Outline

1. Recap and Plan
2. Expectation of a General Random Variable
3. Integration with Respect to a Measure
4. Four Basic Facts
5. Summary and Exercises

What We Have So Far

Two cases of expectation

On $(\Omega, \mathcal{F}, \mathbb{P})$:

1. **Simple** $Y = \sum_{i=1}^{m} y_i I_{B_i}$: $\mathbb{E}[Y] = \sum_i y_i \mathbb{P}(B_i)$.
2. **Non-negative** $X \geq 0$: $\mathbb{E}[X] = \sup\{\mathbb{E}[Y] : Y \text{ simple}, 0 \leq Y \leq X\} \in [0, \infty]$.

An overview of what we will look at here

- Finish expectation: random variables that take **negative** values.
- Then take a break from expectation and study **integration with respect to a measure** μ, written $\int f \, \mathrm{d}\mu$. Expectation will turn out to be the special case $\mu = \mathbb{P}$.

Simple versus discrete

Simple $\Rightarrow$ discrete, but **not** conversely. A Poisson or geometric random variable is discrete and non-negative, but takes infinitely many values, so it is not simple. It is still covered by the supremum definition, because that definition works for *every* non-negative random variable.

Outline

1. Recap and Plan
2. Expectation of a General Random Variable
3. Integration with Respect to a Measure
4. Four Basic Facts
5. Summary and Exercises

Positive and Negative Parts

Definition

For a function (random variable) X define

$$X^+ = \max(X, 0), \qquad X^- = -\min(X, 0) = \max(-X, 0).$$

Properties

- $X^+ \geq 0$ and $X^- \geq 0$.
- $X = X^+ - X^-$ and $|X| = X^+ + X^-$.
- At each ω, at most one of $X^+(\omega), X^-(\omega)$ is nonzero.

Definition of $\mathbb{E}[X]$ for General X

Definition

For a random variable X (any sign), define

$$\mathbb{E}[X] = \mathbb{E}[X^+] - \mathbb{E}[X^-].$$

Both terms are expectations of **non-negative** random variables, which we already know how to define.

The three cases

$\mathbb{E}[X^+]$	$\mathbb{E}[X^-]$	$\mathbb{E}[X]$
finite	finite	the real number $\mathbb{E}[X^+] - \mathbb{E}[X^-]$
$+\infty$	finite	$+\infty$
finite	$+\infty$	$-\infty$
$+\infty$	$+\infty$	**does not exist**

Never write $\infty - \infty = 0$

The two infinities need not be "the same infinity", so the difference has no meaning. In that case we say $\mathbb{E}[X]$ does not exist.

Examples

Example 1 (finite)

X takes $-2, 0, 3$ with probabilities $\frac{1}{3}, \frac{1}{2}, \frac{1}{6}$. From the table: $\mathbb{E}[X^+] = 3 \cdot \frac{1}{6} = \frac{1}{2}$ and $\mathbb{E}[X^-] = 2 \cdot \frac{1}{3} = \frac{2}{3}$, so

$$\mathbb{E}[X] = \tfrac{1}{2} - \tfrac{2}{3} = -\tfrac{1}{6}.$$

Direct check: $(-2)\frac{1}{3} + 0 \cdot \frac{1}{2} + 3 \cdot \frac{1}{6} = -\frac{2}{3} + \frac{1}{2} = -\frac{1}{6}$. ✓

Example 2 (expectation does not exist)

Let $\mathbb{P}(X = 2^k) = \mathbb{P}(X = -2^k) = 2^{-k-1}$ for $k = 1, 2, 3, \ldots$ (these probabilities sum to 1). Then

$$\mathbb{E}[X^+] = \sum_{k \geq 1} 2^k \cdot 2^{-k-1} = \sum_{k \geq 1} \tfrac{1}{2} = \infty, \qquad \mathbb{E}[X^-] = \infty.$$

Positive and negative parts are both infinite, so $\mathbb{E}[X]$ is undefined. By symmetry you might be tempted to say 0, but that is not allowed.

Consistency

If $X \geq 0$, then $X^+ = X$ and $X^- = 0$, so the new definition gives $\mathbb{E}[X] - \mathbb{E}[0] = \mathbb{E}[X]$, the old one.

Are X^+ and X^- Random Variables?

Why check?

The definition only makes sense if X^+ and X^- are **measurable**, so that their expectations are defined. We build the argument from earlier lessons.

Ingredients

- $Y \equiv 0$ is a random variable (inverse image of a Borel set is Ω if it contains 0, else $\varnothing$).
- Lesson 19: $\{X > Y\} \in \mathcal{F}$ for random variables X, Y. Swapping names, $\{Y > X\} \in \mathcal{F}$, so its complement $\{X \geq Y\} = \{Y > X\}^c \in \mathcal{F}$.
- Lesson 20 (gluing): if $A \in \mathcal{F}$ and U, V are random variables, then U on A and V on A^c is a random variable.

Conclusion

With $A = \{X \geq 0\} \in \mathcal{F}$,

$$X^+(\omega) = \begin{cases} X(\omega), & \omega \in A, \\ 0, & \omega \in A^c, \end{cases}$$

is a random variable. The same argument works for X^- (glue $-X$ on $\{X \leq 0\}$ with 0 elsewhere).

Direct alternative

$\{X^+ \leq x\} = \varnothing$ if $x < 0$ and $= \{X \leq x\}$ if $x \geq 0$. Both lie in $\mathcal{F}$.

A Remark on How We Will Compute

Analogy with calculus

In first-year calculus you meet integrals as limits of sums over partitions, but you do not compute them that way. You use rules such as $\int x^n\,dx = \frac{x^{n+1}}{n+1}$. Our definitions of $\mathbb{E}$ (suprema and limits of simple approximations) play the role of the partition definition. Practical rules come later.

Also note

We defined $\mathbb{E}[X] = \mathbb{E}[X^+] - \mathbb{E}[X^-]$ *by definition*, so no linearity is needed to write it. Linearity of $\mathbb{E}$ for general random variables is a theorem to be proved later.

Terminology

X is called **integrable** when $\mathbb{E}[X^+] < \infty$ and $\mathbb{E}[X^-] < \infty$ (equivalently $\mathbb{E}|X| < \infty$, because $|X| = X^+ + X^-$).

Outline

1. Recap and Plan

2. Expectation of a General Random Variable

3. Integration with Respect to a Measure

4. Four Basic Facts

5. Summary and Exercises

Notation and the Idea

Setting

Ω is any nonempty set, $\mathcal{F}$ a σ-field, μ a measure on $\mathcal{F}$. $(\Omega, \mathcal{F})$ is a **measurable space**, and $(\Omega, \mathcal{F}, \mu)$ is a **measure space**. If $\mu(\Omega) = 1$ it is a probability space and we write $\mu = \mathbb{P}$.

Notation (three ways to write the same thing)

$$\int_A f \, \mathrm{d}\mu \;=\; \int_A f(x) \, \mu(\mathrm{d}x) \;=\; \int_A f(x) \, \mathrm{d}\mu(x).$$

Special cases: $\mu = \mathbb{P}$ gives $\int f \, \mathrm{d}\mathbb{P}$; $\mu =$ Lebesgue measure λ gives $\int f \, \mathrm{d}\lambda$.

Idea

In the Riemann integral, the height of a rectangle is f and its base is a **length**. Keep the heights, but measure each base with μ instead. A different μ gives a different kind of "area".

Integral of a Simple Function

Definition

A **simple function** is $f = \sum_{i=1}^{n} a_i I_{A_i}$ with $a_i \geq 0$ and $A_i \in \mathcal{F}$ (measurable). We assume the canonical form: a_i distinct, A_i disjoint and covering Ω. Define

$$\int_{\Omega} f \, \mathrm{d}\mu = \sum_{i=1}^{n} a_i \, \mu(A_i).$$

(Convention: $0 \cdot \infty = 0$. We take $a_i \geq 0$ because we have only defined integrals of non-negative functions so far.)

Over a subset $B \in \mathcal{F}$

$$\int_{B} f \, \mathrm{d}\mu = \sum_{i=1}^{n} a_i \, \mu(A_i \cap B).$$

Picture

Heights are a_1, a_2, a_3; the base under height a_i is $\mu(A_i)$ (even if A_i has several separate pieces). To integrate over B, count only the part of each base lying in B: $\mu(A_i \cap B)$. If $A_i \cap B = \varnothing$ the term is $0 \cdot$ anything $= 0$.

Example: Same Function, Different Measures

Function

$\Omega = [0,1]$, $f = 3\, I_{[0,\frac{1}{4})} + 1\, I_{[\frac{1}{4},1]}$ (so $A_1 = [0,\frac{1}{4})$, $A_2 = [\frac{1}{4},1]$).

Lebesgue measure λ (length)

$$\int_\Omega f \,\mathrm{d}\lambda = 3 \cdot \tfrac{1}{4} + 1 \cdot \tfrac{3}{4} = \tfrac{3}{2}.$$

Over $B = [\frac{1}{8},\frac{1}{2}]$: $\lambda(A_1 \cap B) = \frac{1}{8}$ and $\lambda(A_2 \cap B) = \frac{1}{4}$, so

$$\int_B f \,\mathrm{d}\lambda = 3 \cdot \tfrac{1}{8} + 1 \cdot \tfrac{1}{4} = \tfrac{5}{8}.$$

Dirac measure δ_0 ($\delta_0(A) = 1$ if $0 \in A$, else 0)

Only A_1 contains 0, so $\delta_0(A_1) = 1$ and $\delta_0(A_2) = 0$:

$$\int_\Omega f \,\mathrm{d}\delta_0 = 3 \cdot 1 + 1 \cdot 0 = 3 = f(0).$$

Moral

The **function** f is the same. Only the way we weigh the sets changes.

Lebesgue versus Riemann; Expectation as an Integral

Two ways to cut the area

- **Riemann:** partition the **domain** (the x-axis) into small intervals and choose a height on each (left end, right end, midpoint, ...). If different choices disagree in the limit, the Riemann integral does not exist.
- **Lebesgue:** partition the **range** (the y-axis) and pull each slice back to a set $A_i \subseteq \Omega$; then measure the set with μ.

With Lebesgue measure

Bases are measured by ordinary length, so we recover the usual integral wherever the Riemann integral exists. The Lebesgue integral exists in more cases. Also, Ω need not be numerical, which is why this works for probability spaces.

Expectation is an integral

If $\mu = \mathbb{P}$ and $X = \sum x_i I_{B_i}$ is simple, then

$$\mathbb{E}[X] = \sum_i x_i \mathbb{P}(B_i) = \int_\Omega X \, \mathrm{d}\mathbb{P}.$$

Integral of a Non-Negative Measurable Function

Definition

Let $f : \Omega \to [0, \infty)$ be measurable and $A \in \mathcal{F}$. Define

$$\int_A f \, \mathrm{d}\mu = \sup\Big\{ \int_A s \, \mathrm{d}\mu : \ s \text{ simple, measurable, } 0 \le s \le f \Big\} \in [0, \infty].$$

This is exactly the definition of $\mathbb{E}[X]$ for $X \ge 0$, with $\mu = \mathbb{P}$.

Terminology

- f is **integrable** (with respect to μ) if $\int_\Omega f \, \mathrm{d}\mu < \infty$.
- f is **integrable over** A if $\int_A f \, \mathrm{d}\mu < \infty$.
- If the integral is ∞, we say f is not integrable (or has infinite integral).

Why "over A" matters

For $f \ge 0$ the integral over A is at most the integral over Ω (Fact 2 below). So f can be integrable over a small set A and still not be integrable over all of Ω.

Outline

1. Recap and Plan

2. Expectation of a General Random Variable

3. Integration with Respect to a Measure

4. Four Basic Facts

5. Summary and Exercises

Setting for the Facts

Throughout

$(\Omega, \mathcal{F}, \mu)$ is a measure space, $f, g : \Omega \to [0, \infty)$ are measurable, and $A, B \in \mathcal{F}$.

The facts

1. If $0 \leq f \leq g$, then $\int_A f \, d\mu \leq \int_A g \, d\mu$. (monotone in the function)
2. If $A \subseteq B$, then $\int_A f \, d\mu \leq \int_B f \, d\mu$. (monotone in the set)
3. For a constant $c \geq 0$: $\int_A cf \, d\mu = c \int_A f \, d\mu$.
4. If $f = 0$ on A, then $\int_A f \, d\mu = 0$.

Why prove obvious facts?

They look obvious, but we must check them from the supremum definition. They are the tools for the next big theorems (the monotone convergence theorem).

Fact 1: $0 \le f \le g \Rightarrow \int_A f \, \mathrm{d}\mu \le \int_A g \, \mathrm{d}\mu$

Idea

Every step function that fits under f also fits under g. So the supremum for g runs over **at least** as many step functions, and a supremum over a bigger set is at least as large.

Proof

Let s be simple with $0 \le s \le f$. Since $f \le g$, also $0 \le s \le g$. So

$$\{s : \ 0 \le s \le f\} \ \subseteq \ \{s : \ 0 \le s \le g\}.$$

Hence

$$\int_A f \, \mathrm{d}\mu = \sup_{0 \le s \le f} \int_A s \, \mathrm{d}\mu \ \le \ \sup_{0 \le s \le g} \int_A s \, \mathrm{d}\mu = \int_A g \, \mathrm{d}\mu. \qquad \square$$

Fact 2: $A \subseteq B \Rightarrow \int_A f\,\mathrm{d}\mu \leq \int_B f\,\mathrm{d}\mu$

Idea

For a fixed step function s, integrating over a bigger set counts at least as much base. Pass to the supremum.

Proof

Let $s = \sum_i a_i I_{A_i}$ be simple with $0 \leq s \leq f$. Since $A \subseteq B$, $A_i \cap A \subseteq A_i \cap B$, and by monotonicity of μ,

$$\mu(A_i \cap A) \leq \mu(A_i \cap B) \qquad \text{for each } i.$$

Multiply by $a_i \geq 0$ and sum:

$$\int_A s\,\mathrm{d}\mu = \sum_i a_i\,\mu(A_i \cap A) \;\leq\; \sum_i a_i\,\mu(A_i \cap B) = \int_B s\,\mathrm{d}\mu \;\leq\; \int_B f\,\mathrm{d}\mu.$$

Pass to the supremum

So $\int_B f\,\mathrm{d}\mu$ is an **upper bound** for all the numbers $\int_A s\,\mathrm{d}\mu$. The supremum is the *least* upper bound, hence

$$\int_A f\,\mathrm{d}\mu = \sup_s \int_A s\,\mathrm{d}\mu \leq \int_B f\,\mathrm{d}\mu. \qquad \square$$

Fact 3: $\int_A cf\,\mathrm{d}\mu = c\int_A f\,\mathrm{d}\mu$ for $c \geq 0$

You can do this as an exercise

Here is the proof, following the hints given there. We need $c \geq 0$ only because integrals of negative functions are not yet defined.

Case $c = 0$

$0 \cdot f \equiv 0$. The only simple s with $0 \leq s \leq 0$ is $s \equiv 0$, with integral 0. So $\int_A 0\,\mathrm{d}\mu = 0 = 0 \cdot \int_A f\,\mathrm{d}\mu$ (with $0 \cdot \infty = 0$ if needed).

Case $c > 0$

- s is simple with $0 \leq s \leq f \iff cs$ is simple with $0 \leq cs \leq cf$. So the simple functions under cf are exactly the cs with s under f.
- For simple $s = \sum a_i I_{A_i}$: $\int_A cs\,\mathrm{d}\mu = \sum ca_i\mu(A_i \cap A) = c\int_A s\,\mathrm{d}\mu$.

Hence

$$\int_A cf\,\mathrm{d}\mu = \sup_{s\leq f}\int_A cs\,\mathrm{d}\mu = \sup_{s\leq f} c\int_A s\,\mathrm{d}\mu = c\sup_{s\leq f}\int_A s\,\mathrm{d}\mu = c\int_A f\,\mathrm{d}\mu. \qquad \square$$

Fact 4: $f = 0$ on $A \Rightarrow \int_A f \, \mathrm{d}\mu = 0$

Idea

If f is zero on A, every step function underneath is also zero on A. Over A there is nothing to count, so every integral in the supremum is 0, and so is the supremum.

Proof

Let $s = \sum_i a_i I_{A_i}$ be simple with $0 \leq s \leq f$, in canonical form. We show $\int_A s \, \mathrm{d}\mu = 0$ by checking each term $a_i \mu(A_i \cap A)$:

- $A_i \cap A = \varnothing$: then $\mu(A_i \cap A) = \mu(\varnothing) = 0$, so the term is 0.
- $A_i \cap A \neq \varnothing$ (the measure may still be 0, or not): pick $\omega \in A_i \cap A$. Then $\omega \in A$, so $f(\omega) = 0$, hence $0 \leq s(\omega) \leq 0$. But $s(\omega) = a_i$ since $\omega \in A_i$. So $a_i = 0$ and the term is 0.

Thus every $\int_A s \, \mathrm{d}\mu = 0$, and the supremum of zeros is 0:

$$\int_A f \, \mathrm{d}\mu = 0. \qquad \square$$

A Numerical Look at Fact 4

Example

$\Omega = [0,1]$, Lebesgue measure, and $f = I_{[0,\frac{1}{2}]} \cdot 1$ (so $f = 0$ on $A = (\frac{1}{2}, 1]$). Take a step function underneath f:

$$s = \tfrac{1}{2} I_{[0,\frac{1}{4})} + 0 \cdot I_{[\frac{1}{4},1]} \qquad (0 \le s \le f).$$

Integral over $A = (\frac{1}{2}, 1]$

$$\int_A s \, d\lambda = \tfrac{1}{2}\,\lambda\big([0,\tfrac{1}{4}) \cap (\tfrac{1}{2}, 1]\big) + 0 \cdot \lambda\big([\tfrac{1}{4}, 1] \cap (\tfrac{1}{2}, 1]\big) = \tfrac{1}{2} \cdot 0 + 0 \cdot \tfrac{1}{2} = 0.$$

What happens

The block $A_2 = [\frac{1}{4}, 1]$ does meet A, but its height is 0, forced because $s \le f = 0$ on A. The block with positive height does not meet A at all.

Outline

1. Recap and Plan
2. Expectation of a General Random Variable
3. Integration with Respect to a Measure
4. Four Basic Facts
5. Summary and Exercises

Summary

Key results

Concept	Content
X^+, X^-	$X^+ = \max(X, 0)$, $X^- = -\min(X, 0)$; $X = X^+ - X^-$, $\lvert X\rvert = X^+ + X^-$; both measurable
General $\mathbb{E}[X]$	$\mathbb{E}[X^+] - \mathbb{E}[X^-]$; $\pm\infty$ allowed; undefined if both infinite
Measure space	$(\Omega, \mathcal{F}, \mu)$; $\int_A f \,\mathrm{d}\mu$, $\int_A f(x)\,\mu(\mathrm{d}x)$, $\int_A f \,\mathrm{d}\mu(x)$ are the same
Simple f	$\int_B f \,\mathrm{d}\mu = \sum_i a_i \mu(A_i \cap B)$
$f \geq 0$	$\int_A f \,\mathrm{d}\mu = \sup\{\int_A s \,\mathrm{d}\mu : \ s \text{ simple}, \ 0 \leq s \leq f\}$
Expectation	$\mathbb{E}[X] = \int_\Omega X \,\mathrm{d}\mathbb{P}$
Fact 1	$f \leq g \Rightarrow \int_A f \leq \int_A g$
Fact 2	$A \subseteq B \Rightarrow \int_A f \leq \int_B f$
Fact 3	$\int_A cf = c \int_A f$ $(c \geq 0)$
Fact 4	$f = 0$ on $A \Rightarrow \int_A f \,\mathrm{d}\mu = 0$

Exercises

1. Let X take values $-3, 1, 4$ with probabilities $\frac{1}{4}, \frac{1}{2}, \frac{1}{4}$. Find X^+, X^-, $\mathbb{E}[X^+]$, $\mathbb{E}[X^-]$, $\mathbb{E}[X]$ and $\mathbb{E}|X|$.
2. Show that $X = X^+ - X^-$ and $|X| = X^+ + X^-$ for every real number X.
3. Show directly that X^- is a random variable by gluing $-X$ on $\{X \leq 0\}$ with 0 on $\{X > 0\}$. Which lemma from Lesson 20 do you use?
4. Give an example (different from the lecture's) of a random variable with $\mathbb{E}[X^+] = \infty$, $\mathbb{E}[X^-] < \infty$. What is $\mathbb{E}[X]$?
5. With $\Omega = [0, 1]$ and $f = 2I_{[0,\frac{1}{3})} + 5I_{[\frac{1}{3},1]}$, compute $\int f \, \mathrm{d}\lambda$ and $\int_{[0,\frac{1}{2}]} f \, \mathrm{d}\lambda$.
6. Let μ be the **counting measure** on $\Omega = \{1, 2, 3\}$ with all subsets measurable, and $f(1) = 2$, $f(2) = 5$, $f(3) = 7$. Compute $\int f \, \mathrm{d}\mu$.
7. For the Dirac measure δ_{x_0}, show that $\int_\Omega f \, \mathrm{d}\delta_{x_0} = f(x_0)$ for simple f.
8. In the proof of Fact 2, where is monotonicity of the measure used? In the proof of Fact 4, where is $\mu(\varnothing) = 0$ used?
9. Show that $\int_\Omega f \, \mathrm{d}\mu = 0$ and $f \geq 0$ together do **not** imply $f = 0$ everywhere. (Hint: take μ to be Lebesgue measure and $f = I_{\{0\}}$.)

Outlook

Next lesson

A few more basic facts about $\int f \, \mathrm{d}\mu$ for non-negative f, then **Lebesgue's monotone convergence theorem**: if $0 \leq f_n \uparrow f$, then $\int f_n \, \mathrm{d}\mu \uparrow \int f \, \mathrm{d}\mu$.

Looking ahead

This is the exact analogue of what we proved for $\mathbb{E}[X_n] \uparrow \mathbb{E}[X]$ in Lesson 25. The question for next time is whether that proof used anything special about a *probability* measure, or whether it works for every measure μ.